

# Matrix DAE Analysis of Inverter-Dominated Grids: Kronecker Assembly, the Resolvent Operator, and Non-Modal Weak Nodes

A complete step-by-step derivation

## Abstract

This note develops, from first principles, the matrix differential–algebraic (DAE) model of an inverter-dominated grid; assembles it compactly with the Kronecker product (stating precisely *when* that buys real structure and when it does not); builds the *resolvent operator* of the resulting non-normal system; and derives the final, original results: an input–output amplification curve that eigenvalues cannot see, a frequency-resolved Non-Modal Weakness Index, and a transient-aware hosting certificate. Each block is tagged [STD] (standard/reusable) or [NEW] (contribution).

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## 1 Step 0 — Notation and objects [STD]

Graph  $G = (V, E, W)$ ,  $|V| = n$ , connected. Per node  $i$ : phase  $\theta_i$ , frequency deviation  $\omega_i$ , inertia  $M_i > 0$ , damping  $D_i \geq 0$ . Stacked vectors  $\theta, \omega \in \mathbb{R}^n$ . Diagonal operators  $M = \text{diag}(M_i)$ ,  $D = \text{diag}(D_i)$ . Weighted Laplacian  $L = L^\top \succeq 0$ ,  $L\mathbf{1} = 0$ ,  $w_{ij} = B_{ij}|V_i||V_j|$ .  $I_k$  is the  $k \times k$  identity;  $\otimes$  the Kronecker product;  $\text{vec}(\cdot)$  the column-stacking operator. Key identity used throughout:

$$(B^\top \otimes A) \text{vec}(X) = \text{vec}(AXB). \quad (1)$$

## 2 Step 1 — The full nonlinear DAE [STD]

Partition buses into dynamic  $V_d$  and algebraic  $V_\ell$ . The structure-preserving model of the AC grid is the semi-explicit DAE

$$\dot{\theta}_d = \omega_d, \quad (2a)$$

$$M\dot{\omega}_d = -D\omega_d + P_d - P_e(\theta, |V|), \quad (2b)$$

$$0 = P_\ell - P_{e,\ell}(\theta, |V|), \quad (2c)$$

$$0 = Q_\ell - Q_{e,\ell}(\theta, |V|), \quad (2d)$$

with the network injection operator (Kirchhoff-exact)

$$P_e(\theta, |V|) = \text{diag}(|V|) [(G \circ \cos \Theta + B \circ \sin \Theta) |V|], \quad \Theta_{ij} = \theta_i - \theta_j. \quad (3)$$

Equations (2) are the object every downstream step linearizes.

## 3 Step 2 — Linearization and the per-node state block [STD]

Linearize (2) at an operating point  $(\theta^*, \omega^*)$  with stiff voltages  $|V| \approx \mathbf{1}$ , high  $X/R$ . The active-power Jacobian  $\partial P_e / \partial \theta$  evaluated at  $\theta^*$  is the weighted Laplacian  $L$ . Define the per-node state  $x_i = [\theta_i, \omega_i]^\top \in \mathbb{R}^2$  and the *local* (isolated) dynamics

$$\dot{x}_i = A_{\text{loc},i} x_i, \quad A_{\text{loc},i} = \begin{bmatrix} 0 & 1 \\ 0 & -\gamma_i \end{bmatrix}, \quad \gamma_i = \frac{D_i}{M_i}. \quad (4)$$

Coupling enters only through the lower-left block: a phase difference at neighbours produces a torque on  $\dot{\omega}_i$ . Encode *which* local variable is exchanged with the coupling selector

$$\Gamma = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 2}. \quad (5)$$

## 4 Step 3 — Kronecker assembly: when it pays [STD→NEW]

Stack  $X = [x_1^\top, \dots, x_n^\top]^\top \in \mathbb{R}^{2n}$  (node-major ordering). The global linear system is

$$\dot{X} = \left[ I_n \otimes A_{\text{loc}} - (M^{-1}L) \otimes \Gamma \right] X + (I_n \otimes B_0) u \quad (6)$$

with  $B_0 = [0, 1]^\top$  injecting power into the frequency channel,  $u \in \mathbb{R}^n$  the nodal disturbance. The first term is purely local (block-diagonal); the second is purely topological. *This is the entire value of Kronecker here:* it cleanly separates device physics ( $A_{\text{loc}}$ ) from network ( $M^{-1}L$ ).

**Remark 1** (When Kronecker *stops* helping — read this). Equation (6) factors as a clean Kronecker *sum* only because every node shares the *same*  $A_{\text{loc}}$  structure and  $\gamma_i$  enters diagonally. The eigenvalues would then be the tidy set  $\lambda(I \otimes A_{\text{loc}} - (M^{-1}L) \otimes \Gamma)$ . **But the moment nodes are heterogeneous** (SG vs. GFM vs. GFL, i.e.  $\gamma_i$  differ *and* GFL adds a frequency→angle feedback), the two factors *do not commute*:

$$[I \otimes A_{\text{loc}}][(M^{-1}L) \otimes \Gamma] \neq [(M^{-1}L) \otimes \Gamma][I \otimes A_{\text{loc}}]. \quad (7)$$

Non-commuting Kronecker factors  $\Rightarrow$  the spectrum is *not* the Kronecker sum of the factor spectra, and the eigenvectors are *not* orthogonal. This is the precise algebraic origin of non-normality. **Conclusion: Kronecker is worth using for the *assembly* and for exposing non-commutation as the source of fragility — but it is *not* worth using to “diagonalize” the system, because it cannot.** We therefore stop the Kronecker route here and switch to the resolvent (Step 6).

## 5 Step 4 — Mass-normalized reduction [STD]

Re-order to state-major  $x = [\theta^\top, \omega^\top]^\top$  (equivalent to (6) by a permutation). Then

$$\dot{x} = A_{\text{sys}}x + Bu, \quad A_{\text{sys}} = \begin{bmatrix} 0 & I_n \\ -M^{-1}L & -M^{-1}D \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ M^{-1} \end{bmatrix}. \quad (8)$$

Remove the center-of-inertia null mode ( $L1 = 0$ ) and apply the inertial isometry  $\vartheta = M^{1/2}\tilde{\theta}$ :

$$\ddot{\vartheta} + \tilde{D}\dot{\vartheta} + \tilde{L}\vartheta = M^{-1/2}\tilde{u}, \quad \tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (9)$$

Spectrum  $0 = \nu_1 < \nu_2 \leq \dots \leq \nu_n$ ;  $\nu_c := \nu_2(\tilde{L})$ .

## 6 Step 5 — The eigenvalue certificate and its hidden hypothesis [STD]

Modal poles (weak coupling  $\tilde{D} \approx \gamma I$ ):

$$s_{k,\pm} = -\frac{\delta_k}{2} \pm \sqrt{\left(\frac{\delta_k}{2}\right)^2 - \nu_k}, \quad \delta_k = q_k^\top \tilde{D} q_k. \quad (10)$$

Certificate:  $\nu_k > 0, \delta_k > 0 \forall k$ . **Hidden hypothesis:** (10) governs the *response* only if  $A_{\text{sys}}$  is normal,  $A_{\text{sys}}^\top A_{\text{sys}} = A_{\text{sys}} A_{\text{sys}}^\top$ , i.e.  $\|e^{A_{\text{sys}}t}\| = e^{t\alpha(A_{\text{sys}})}$ ,  $\alpha = \max_k \Re \lambda_k$ . By Remark 1, heterogeneity breaks this.

## 7 Step 6 — The resolvent operator [NEW core]

Instead of the homogeneous problem  $\dot{x} = A_{\text{sys}}x$ , study the *forced* problem in the frequency domain. Take  $u(t) = \hat{u} e^{i\Omega t}$ ; the steady response is  $x(t) = \hat{x} e^{i\Omega t}$  with

$$\hat{x} = \underbrace{(i\Omega I - A_{\text{sys}})^{-1} B}_{\mathcal{R}(\Omega)} \hat{u}. \quad (11)$$

$\mathcal{R}(\Omega)$  is the **resolvent** (transfer function). Restrict the output to the frequency channel via  $C = [0 \ I_n]$  and define the input–output operator  $H(\Omega) = C\mathcal{R}(\Omega)$ . Its singular value decomposition is the whole story:

$$H(\Omega) = \sum_j \sigma_j(\Omega) \hat{r}_j(\Omega) \hat{f}_j(\Omega)^* \quad (12)$$

where, at each frequency  $\Omega$ :

- $\sigma_1(\Omega) \geq \sigma_2 \geq \dots \geq 0$  are the **gains**;
- $\hat{f}_1(\Omega)$  (right singular vector) is the **optimal forcing** — the worst-case spatial disturbance pattern;
- $\hat{r}_1(\Omega)$  (left singular vector) is the **optimal response** — where the amplified frequency swing appears.

**Proposition 1** (Normal vs. non-normal gain). *If  $A_{\text{sys}}$  is normal,  $\sigma_1(\Omega) = (\min_k |i\Omega - \lambda_k|)^{-1}$ : the gain peaks only at the eigenfrequencies and is fully predicted by the spectrum. If  $A_{\text{sys}}$  is non-normal,  $\sigma_1(\Omega)$  can exceed this by the condition number of the eigenvector basis, peaking at frequencies and amplitudes no eigenvalue predicts.*

*Proof.* Normal  $A_{\text{sys}} = U\Lambda U^*$  with unitary  $U$  gives  $H(\Omega) = CU(i\Omega I - \Lambda)^{-1}U^*B$ , a unitary conjugation of a diagonal operator, whose largest singular value is the largest diagonal magnitude  $\max_k |i\Omega - \lambda_k|^{-1}$ . Non-normal  $A_{\text{sys}} = V\Lambda V^{-1}$  with non-unitary  $V$  inserts  $\kappa(V) = \|V\| \|V^{-1}\| > 1$ , so  $\sigma_1(\Omega) \leq \kappa(V) \max_k |i\Omega - \lambda_k|^{-1}$  with the bound active away from eigenfrequencies.  $\square$

## 8 Step 7 — Three norms, one operator [NEW]

The resolvent unifies every margin in the paper:

$$\text{Worst-case sustained gain } (H_\infty): \quad \gamma_\infty = \sup_{\Omega} \sigma_{\max}(H(\Omega)); \quad (13)$$

$$\text{Stochastic variance } (H_2): \quad \|H\|_2^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \|H(\Omega)\|_F^2 d\Omega = \text{Tr}(\tilde{L}^\dagger) \cdot \text{const}; \quad (14)$$

$$\text{Worst-case transient (Kreiss):} \quad \mathcal{K} = \sup_{\Re z > 0} \Re z \left\| (zI - A_{\text{sys}})^{-1} \right\|. \quad (15)$$

Equation (14) *recovers* the Spectral Federation / variance result as the integral of the resolvent gain — i.e. federation is the area under the gain curve, while  $H_\infty$  is its peak. **Pseudospectrum and resolvent are the same object viewed two ways:**  $\sigma_\varepsilon(A_{\text{sys}}) = \{z : \|(zI - A_{\text{sys}})^{-1}\| \geq \varepsilon^{-1}\}$  is the level set of the resolvent norm in the complex plane;  $\sigma_1(\Omega)$  is its trace along the imaginary axis.

## 9 Step 8 — Result A: the amplification curve [NEW, the wow figure]

Sweep IBR penetration  $\rho \in [0, 1]$  (convert SG→GFL:  $M_i = M_i^0(1 - \rho) + \rho\epsilon$ ,  $D_i$  from PLL). Plot  $\sigma_1(\Omega; \rho)$ .

- High inertia ( $\rho \rightarrow 0$ ): flat, low curve — the grid rejects all forcing.
- Inverter-dominated ( $\rho \rightarrow 1$ ): a sharp resonant peak emerges at the inter-area frequency  $\Omega \approx \sqrt{\nu_c}$ , with height  $\gg$  the eigenvalue prediction (Prop. 1).

**This single plot shows a grid that is eigenvalue-stable ( $\nu_c > 0$ ) yet has a large-gain window where a modest disturbance is amplified past protection thresholds.**

## 10 Step 9 — Result B: frequency-resolved Non-Modal Weak Nodes [NEW]

Read the optimal forcing at the peak frequency  $\Omega^* = \arg \max_{\Omega} \sigma_1$ :

**Definition 1** (Non-Modal Weakness Index).

$$W_i^{\text{nm}} = \left| [\hat{f}_1(\Omega^*)]_i \right|^2, \quad (\text{response hot-spot}) \quad W_j^{\text{nmout}} = \left| [\hat{r}_1(\Omega^*)]_j \right|^2. \quad (16)$$

**Proposition 2** (Forcing  $\neq$  response  $\neq$  modal  $\neq$  SCR). *For non-normal  $A_{\text{sys}}$ , the four rankings  $W^{\text{nm}}$  (where to inject),  $W^{\text{nmout}}$  (where it hurts),  $q_{c,i}^2/M_i$  (modal participation), and SCR (steady-state) are in general distinct. In particular the peak input node and peak output node need not coincide.*

*Sketch.*  $\hat{f}_1, \hat{r}_1$  are the leading right/left singular vectors of  $H(\Omega^*)$ ; they coincide only when  $H$  is normal. Misalignment of  $\hat{f}_1$  and  $\hat{r}_1$  is exactly the non-normal “input here, output there” effect, absent from any eigenvector-based metric.  $\square$

## 11 Step 10 — Result C: Dynamic Braess on the gain curve [NEW]

Reinforce a line,  $w_{ij} \rightarrow w_{ij} + \Delta$ . Then  $\nu_c \uparrow$  (Rayleigh), so the eigenvalue certificate improves, yet the GFL interaction raises  $\kappa(V)$  and hence the resolvent peak:

$$\frac{\partial \nu_c}{\partial w_{ij}} > 0 \quad \text{while} \quad \frac{\partial}{\partial w_{ij}} \left( \sup_{\Omega} \sigma_1(\Omega) \right) > 0. \quad (17)$$

**Braess becomes visible:** the gain curve rises after a “beneficial” reinforcement. Certify reinforcements by  $\gamma_{\infty}$ , not by  $\nu_c$ .

## 12 Step 11 — Result D: transient-aware hosting certificate [NEW]

Let  $\Gamma_{\text{trip}}$  be the protection amplification margin. Define

$$\boxed{\rho_{\text{io}}^* = \max \left\{ \rho : \gamma_{\infty}(\rho) = \sup_{\Omega} \sigma_{\max}(H(\Omega; \rho)) \leq \Gamma_{\text{trip}} \right\}} \quad (18)$$

Computed by a 1-D sweep over  $\Omega$  at each  $\rho$  (cheap; no time-domain, no Monte Carlo).

**Proposition 3.**  $\rho_{\text{io}}^* \leq \rho_{\text{eig}}^*$ : the eigenvalue hosting limit over-grants capacity by the shaded gap; the gap widens monotonically with the departure from normality  $d_F(A_{\text{sys}}(\rho))$ .

## 13 Step 12 — Computation recipe (IEEE 39-bus) [STD]

1. Build  $A_{\text{sys}}(\rho)$  from (8); convert SG at buses 30–39 to GFL with realistic low-bandwidth PLL.
2. For a grid of  $\Omega$ : form  $H(\Omega) = C(i\Omega I - A_{\text{sys}})^{-1}B$ , take its SVD, store  $\sigma_1, \hat{f}_1, \hat{r}_1$ .
3. **Fig. A:**  $\sigma_1(\Omega; \rho)$  surface (the wow plot).
4. **Fig. B:** bar maps of  $\hat{f}_1$  vs.  $\hat{r}_1$  at  $\Omega^*$  — circle the input/output mismatch.
5. **Fig. C:** overlay eigenvalues (all LHP) with  $\sigma_{\varepsilon}$  crossing into RHP; confirm  $\sigma_1$  peak coincides with pseudospectral protrusion.
6. **Fig. D:**  $\rho_{\text{eig}}^*$  vs.  $\rho_{\text{io}}^*$  gap vs.  $d_F$ .

## 14 Reusable vs. original ledger

| Ingredient                                                               | Origin                            |
|--------------------------------------------------------------------------|-----------------------------------|
| Structure-preserving DAE, Laplacian, $\nu_c$                             | [STD] Kundur; Dörfler; your draft |
| Kronecker assembly $I \otimes A_{\text{loc}} - (M^{-1}L) \otimes \Gamma$ | [STD] standard MAS algebra        |
| Non-commutation $\Rightarrow$ non-normality (this framing)               | [NEW]                             |
| Resolvent SVD as grid amplification operator                             | [NEW] (tool [STD] from fluids)    |
| Frequency-resolved Non-Modal Weakness (input $\neq$ output)              | [NEW]                             |
| $H_2$ = area, $H_\infty$ = peak of one gain curve                        | [NEW unification]                 |
| Braess on the gain curve; I/O hosting certificate                        | [NEW]                             |

## 15 Honest caveats

1. Classical resolvent assumes asymptotic stability ( $\nu_c > 0$ ); beyond the hosting limit use the *discounted* resolvent  $(i\Omega + \beta)I - A_{\text{sys}}$ .
2.  $H_\infty$  is known in power-system control; the novelty is the *modal* reading of the resolvent (singular vectors as frequency-resolved weak nodes) for IBR hosting.
3. Verify the  $\sigma_1$  peak *separates* the inverter case from the synchronous case in *your* 39-bus before claiming it; idealized PLLs flatten the peak.
4. Kronecker is used for assembly and to expose non-commutation only — do not claim a Kronecker-sum spectral solution; it does not exist here.